Where Cross-Question Structure in LLM Errors Comes From: Shared Seeds, Shared Prompts, and Answer-Letter Preference

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Abstract

When language models disagree on a question, the disagreement is often read as missing knowledge. Questions that need the same missing knowledge should then be answered right or wrong together. We test for this structure with the eigenvalues of the questions' correlation matrix. Within one model, independently drawn answers show none that the test can detect; it appears when all questions in one pass over the benchmark share a random seed or a few-shot prompt. Across 24 models, the largest structure beyond overall skill is a preference among answer letters: when the options are rotated, it follows the letter, not the question, and it depends on one model, Llama-3.2-1B, without which only a smaller excess remains. Counting a question correct only if it is right in every rotation removes the part that acts on guesses. What remains is a marginal excess at the 5% line, which a post hoc analysis ties to two pairs of near-duplicate models.

1. Introduction

Sampling-based methods such as self-consistency, SelfCheckGPT and semantic entropy sample a language model several times on the same question and read disagreement among the answers as uncertainty about the answer (Wang et al., 2023; Manakul et al., 2023; Farquhar et al., 2024). In a deep ensemble, disagreement among the members is read as epistemic uncertainty: the part that comes from missing knowledge, rather than from noise that would remain even if the model were known (Lakshminarayanan et al., 2017; Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021). Recent scores add disagreement between models to self-consistency and treat it as epistemic (Hamidieh et al., 2026).

This reading makes a prediction. A model that lacks some piece of knowledge should fail every question that needs it, so across an ensemble those questions should be right or wrong together. We call that tendency *cross-question structure*. Related structure is already documented: different models often choose the same wrong answer (Kim et al., 2025), and a few latent factors summarize benchmark scores (Burnell et al., 2023; Ruan et al., 2024; Maimon et al., 2026). We ask whether cross-question structure really comes from shared knowledge.

Two other sources can produce the same structure, and both are easy to miss. The first is the evaluation pipeline. Within one model, knowledge is fixed, so each question has the same chance of a correct answer on every pass (pass k is the model's kth answer to every question, not pass@k). Knowledge therefore cannot make two questions right on the same passes and wrong on the others; only something the questions share inside a pass can. A shared random seed is one example. All questions in the pass then receive the same random numbers from the sampler, so a draw that favors the letter "B" at one generation step favors it in every question whose answer letter is emitted at that step. A shared few-shot prompt is another, since few-shot examples shape answers (Zhao et al., 2021; Min et al., 2022).

The second source is a letter preference of the model. Llama-3.2-1B is accurate when the correct answer is placed at "A", and below the 0.25 of a random guess when it is placed at the other letters (Fig. 1). It answers "A" in 81% of its replies in the original option order, and a know-or-guess model fitted to its answers (Sec. 2) puts 0.90 of its guesses on "A" (App. D). Questions whose correct answer is "A" then look like questions it knows, and across an ensemble those questions are right or wrong together. That is the pattern shared missing knowledge would produce. Letter preferences of single models (also called selection bias) are already documented (Zheng et al., 2024; Pezeshkpour & Hruschka, 2024; Ding, 2026). Rotating the options separates a letter preference from a knowledge gap. A knowledge gap stays with the question. A letter preference stays with the answer's letter.

We test the two sources separately. Within one model, we compare independent draws with passes that share a seed or

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2nd Workshop on Epistemic Intelligence in Machine Learning (EIML@ICML 2026), Seoul, South Korea. *Copyright 2026 by the author(s).*

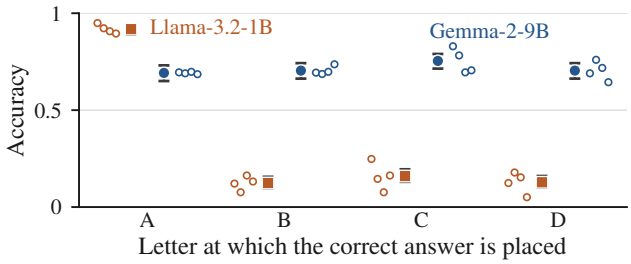


Figure 1. Llama-3.2-1B is accurate where the answer is placed at "A", in each of the four rotations; Gemma-2-9B is about equally accurate at every letter. Filled: all 500 questions, with the options rotated so that the correct answer is at that letter; bars: Wilson 95% CI. Open: each of the four rotations alone (about 125 questions).

a few-shot prompt (Sec. 3.1). Across 24 models, we rotate the options to new letters, so the gold letter (the letter of the correct option) changes and the question and options do not (Sec. 3.2), and we count a question correct only when the model is right in every rotation (*all-4-correct*; Sec. 3.3). All three tests use the *correctness matrix*: 1 if an answer is right, 0 if wrong. Structure is an eigenvalue of the correlation matrix of its questions that is larger than a null model allows: independent answers within one model, and, across models, one skill per model (a unidimensional item response model). Spectral summaries of benchmark dimensionality exist (Sha & Zhao, 2026). We add that the correctness matrix records no letters, yet the second eigenvector of its correlation matrix follows a letter preference, and removing one model brings its eigenvalue within the range of one skill (Sec. 3.2).

2. A spectral test for cross-question structure

We record the answers in a correctness matrix $\mathbf{C}$. Entry c_{ki} is 1 if row k answers question i correctly and 0 otherwise. Within one model a row is a pass; across models a row is a model. We keep only questions that vary: within a model, the N_b *borderline* questions, whose pass rate f_i (the share of passes that answer question i correctly) lies between $\varepsilon = 0.05$ and $1 - \varepsilon$; across models, the *split* questions, which some models answer correctly and others do not.

Let $\mathbf{R}$ be the matrix of Pearson (phi) correlations between the 0/1 columns of the kept questions. If the questions were independent, every true correlation would be 0 and every eigenvalue would be 1. With a finite number of rows, the measured correlations are noisy, and the noise spreads the eigenvalues around 1. Marchenko and Pastur showed that the eigenvalues then fill a range whose upper edge is $\lambda_+ = (1 + \sqrt{\gamma})^2$, where γ is the number of questions divided by the number of rows (Marčenko & Pastur, 1967); as both numbers grow, the largest eigenvalue of such a matrix of independent entries converges to that edge (Bai & Silverstein, 2010; Jiang, 2004). More questions per row give a higher edge. For Llama-3-8B's 173 borderline questions and 100 passes, $\gamma = 1.73$ and $\lambda_+ = 5.36$. An eigenvalue above that edge is evidence that some group of questions is right or wrong together more often than noise allows. Its eigenvector gives each question's *loading*, how strongly it belongs to the group, and each row's *score*, its projection.

Within one model, knowledge is fixed, so only the pipeline can make questions right or wrong together (Sec. 1). This test checks the pipeline, not knowledge, and its null model is independence. We count the eigenvalues above the Marchenko–Pastur (MP) edge λ_+ (the *MP count*). Because λ_+ is a limit for large matrices, we also compare how far the largest eigenvalue lies from λ_+, in units of its typical fluctuation, with the same distance in 1000 simulated matrices of the same size, in which each question keeps its pass rate and answers are drawn independently (App. B.1).

Across models, independence is the wrong null model. A strong model is right on many questions and a weak one on few, so any two questions are correlated through skill alone; this gives one large eigenvalue, λ_1. The null model is a one-skill item response model. Model m has skill θ_m, question i has difficulty b_i and discrimination (slope) a_i, and the probability of a correct answer is $\sigma(a_i(\theta_m - b_i))$, with σ the logistic function. This is the two-parameter logistic (2PL) model (Lord & Novick, 1968). With 0/1 answers, skill alone can also produce a second large eigenvalue, the *difficulty factor* of binary items (Ferguson, 1941; McDonald & Ahlawat, 1974), because a 0/1 question correlates more strongly with questions of similar difficulty. In the all-4-correct matrix of Sec. 3.3 (391 split questions, 24 models), $\lambda_+ = 25.4$, yet one-skill simulations have a median λ_2 of 28.2. We fit the model, simulate datasets from it, and say that λ_j, the jth largest eigenvalue (rank j), *exceeds* the null when it lies above the 99th percentile of its simulations.

Two nulls ignore guessing: the 2PL, and Rasch, which gives every question the same discrimination (Rasch, 1960). Because a model that does not know the answer is still right a quarter of the time, we also fit the three-parameter logistic (3PL), with its guessing parameter, the lowest chance of a correct answer, fixed at 0.25, and knowledge-or-random-guessing (*know-or-guess*) nulls. In the uniform one, KG-u, a model that does not know the answer guesses one of the four letters with equal probability (App. D). A null is *adequate* when the observed λ_1, the skill eigenvalue, falls between that null's 1st and 99th percentiles; a null that cannot reproduce skill cannot judge what lies beyond it.

Structure beyond skill can appear in λ_2, λ_3 or λ_4. The *max-statistic test* tests all three at once: it takes the largest of their percentiles among simulations of the same rank and compares it with that maximum in each simulation. Its p-value is p_M.

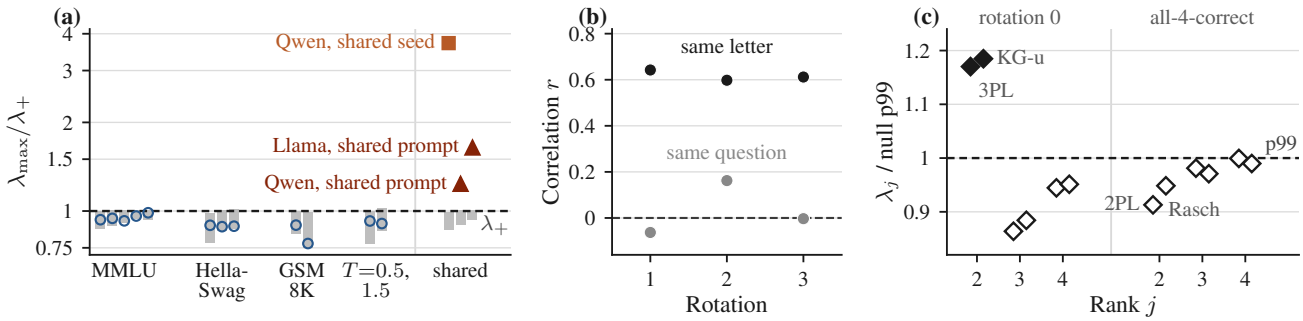


Figure 2. **(a)** Within one model: $\lambda_{\max}/\lambda_+$ (log axis) of each run; grey bar: 5–95% of its simulated z, on this axis; circles: independent draws; filled: an eigenvalue above λ_+. **(b)** Across models, rotations 1–3: correlation over questions of each question's eigenvector-2 loading with the mean rotation-0 loading of the questions that had its new gold letter (same letter), or with its own rotation-0 loading (same question). **(c)** λ_2–λ_4 of the 24-model matrix in rotation 0 and all-4-correct, divided by the 99th percentile (1000 draws) of two adequate one-skill nulls (rotation 0: 3PL left and KG-u, uniform know-or-guess, right; all-4-correct: 2PL left, Rasch right); filled: above it. Other nulls: App. D (soft-preference: not rejected).

Rotations. Each model answers every question four times, with the options moved zero (rotation 0, the original order), one, two or three letters later, cyclically. We count a question correct only if it is right in all four rotations: all-4-correct, the CircularEval rule of Liu et al. (2024). A guess with any fixed letter preference is then correct with probability at most $1/256$ on every question, because the correct answer visits each letter once and the product of the four letter probabilities is at most $(1/4)^4$.

Runs. Within one model there are twelve *independent-draw runs* on MMLU (Hendrycks et al., 2021), HellaSwag (Zellers et al., 2019) and GSM8K (Cobbe et al., 2021), in which each answer has its own seed, as in self-consistency (Table A1). Each run has $K = 30$ to 100 passes. One *shared-seed run* gives all questions of a pass one seed, and two *shared-prompt runs* draw five MMLU development questions (the few-shot *exemplars*) for each pass and place them, with their answers, before every question of that pass. Across models, 24 models from ten families answer the 500 MMLU questions in all four rotations by greedy decoding. A reply that cannot be read as a single answer is *unparseable* and scored wrong; across models, 217 of the 500 questions have at least one such reply (App. A).

3. Results

3.1. Within one model: detectable structure only from a shared seed or prompt

Independent draws show no cross-question structure that the test detects. In all 12 independent-draw runs the MP count is zero, and the distance of the largest eigenvalue from λ_+ lies between the 5th and 95th percentiles of its simulations (Fig. 2a; App. B.1). The test detects weak structure spread over many questions but not a small cluster. A factor shared by the questions of a pass, with latent correlation 0.05 (0/1 correlation about 0.025), is detected in every simulation at Llama-3-8B's 173 borderline questions and in 47% at Qwen2.5-7B's 43. A factor on only nine questions, picked at random or from one subject, is detected in at most a quarter of simulations, even at latent correlation 0.2 (App. B.2).

Sharing a seed or a prompt among the questions of a pass creates structure. In all three such runs the largest eigenvalue lies far above every simulation, and the MP count is two to four. With a shared seed, the cause is the sampler's shared random numbers (Sec. 1; App. C), and in Qwen2.5-7B's run the gold letter explains most of the variance of the top eigenvector's loadings ($\eta^2 = 0.73$, permutation $p = 0.0002$). With a shared prompt, Llama-3-8B copies the exemplars' answer letters, and its top eigenvector also groups by gold letter ($\eta^2 = 0.53$). Qwen2.5-7B does not copy them, and its top eigenvector does not group by gold letter; instead, passes that share an exemplar agree more on borderline questions than passes that share none ($p = 0.0005$). Within one model, each detected structure thus comes from the pipeline, not from knowledge.

3.2. Across models: the second eigenvector follows the answer letter

In rotation 0, the first eigenvector of the 24-model matrix is skill: the models' scores on it correlate with their accuracy at $r = 0.94$. Three nulls with guessing are adequate: the 3PL, KG-u, and the letter-aware null, a know-or-guess null in which a guessing model uses its fitted letter preference. λ_2, 49.9 on 460 split questions, exceeds the 99th percentile of all three (Fig. 2c), no other eigenvalue does, and the max-statistic test rejects them ($p_M = 0.002$).

The second eigenvector follows the answer letter, not the question. Its loadings group by gold letter, and the models' scores on it rise with their share of "A" answers ($r = 0.76$) and fall with their share of "D" answers ($r = -0.90$).

Within one rotation, letter and content cannot be told apart; across rotations they can. Suppose a question's correct answer moves from "B" in rotation 0 to "C" in rotation 1. If its loading comes from content, it matches the question's own rotation-0 loading; if from a letter preference, it matches the mean rotation-0 loading of the questions whose answer was at "C". In rotations 1–3 the loadings follow the letter ($r = 0.60$ to 0.64) and barely follow the question ($r = -0.06$ to 0.16; Fig. 2b). The second eigenvector is a letter effect, not a content effect.

A fitted letter preference accounts for part of this effect, not all of it. The *soft-preference* null, a fourth adequate null, lets that preference act on every answer, not only on guesses. Against it, the max-statistic test does not reject ($p_M = 0.079$), but λ_2 alone is still unlikely ($p = 0.010$), and the null misses how often models are right in exactly one or all four rotations (App. D).

One model, Llama-3.2-1B, lifts λ_2 past the 3PL null. Removing it lowers λ_2 from 49.9 to 35.0, below the 99th percentile of the 3PL null on the other 23 models; removing any other single model leaves it between 47.1 and 52.4. Without it, some structure remains: the letter grouping moves to eigenvector 3 ($\eta^2 = 0.21$), and the max-statistic test still rejects the 3PL null through λ_4 ($p_M = 0.003$; App. D).

3.3. After rotation: a marginal excess tied to a few models

All-4-correct cancels a letter preference that acts only on guesses. In the all-4-correct matrix (391 split questions), λ_2 lies below the 99th percentile of both adequate one-skill nulls, the 2PL and Rasch (Fig. 2c). Because the difficulty factor lifts the nulls' λ_2 above λ_+, a second skill could hide below their 99th percentile. A second skill as strong as the first, planted on the 125 kept questions from STEM subjects, is detected by the max-statistic test in more than 90% of datasets; one half as strong, in about a fifth (App. D).

What remains is a marginal excess on λ_3 and λ_4. The max-statistic test rejects both nulls ($p_M = 0.026$ for the 2PL, 0.041 for Rasch), yet no eigenvalue exceeds the 99th percentile of both (Table A2). The test also rejects slightly too often under one skill (up to 8.7% of datasets at the 5% level); recalibrated, its four p_M run from 0.017 to 0.057, and dropping the 217 questions with an unparseable reply ends the rejection. The excess sits at the 5% line.

The excess is tied to a few models. Before running the max-statistic test, we listed the *sibling groups*, successive releases or fine-tunes in one family at one size: Phi-3/3.5-mini, Llama-3/3.1-8B, Qwen2/2.5-7B, and Mistral-7B v0.2/v0.3 with Zephyr-7B. Keeping one model from each group, in any of the 24 ways, ends the rejection, while removing five other models at random keeps it in all 24 draws. A post hoc follow-up narrows it to two pairs: removing one of Phi-3-mini and Phi-3.5-mini and one of Llama-3-8B and Llama-3.1-8B raises p_M above 0.2 under both nulls, which none of 190 removals of two other models does. Llama-3.2-1B also matters: removing it alone ends the rejection ($p_M = 0.059$ for the 2PL, 0.089 for Rasch), though 12 of the 19 removals of it with another model outside the pairs still reject under the 2PL. Part of its letter preference may remain after all-4-correct, if the preference also tilts answers it partly knows; no adequate null here can test this. Simple baselines agree: Llama-3.2-1B has the most skewed letter shares, and 2PL residual correlations rank the two pairs first and second of 276 (App. D).

4. Discussion

Cross-question structure in correctness is not, by itself, evidence of shared missing knowledge. In our data it came from the pipeline within a model and, in its largest part, from a letter preference across models, and what remained after rotation is marginal and tied to a few models. A letter preference, even one carried by a single model, can create an eigenvector that an ensemble would read as missing knowledge. Scores that treat cross-model disagreement as epistemic (Hamidieh et al., 2026) should be checked for letter preference. A shared seed leaves the distribution of each question's answers unchanged but feeds the same random numbers to every question in the pass, so a comparison across questions can detect it even when the number correct per pass shows nothing.

The excess is marginal (Sec. 3.3). The 217 questions with an unparseable reply may carry part of it, though dropping them also leaves 236 split questions instead of 391. The rule for an adequate null, the single-model and pair removals that isolate the two pairs, and the analyses of test size, power and the difficulty factor were chosen after seeing the data (App. D). Why the models in a pair fail together was not tested; shared knowledge gaps from common training data, contamination and other causes remain open. In the ensemble, architecture, family, scale and training data vary together. The test can also miss structure carried by a few models, and anything outside the correctness matrix, such as log-probabilities. The within-model runs use instruction-tuned models of 1.7B to 8B, and the across-model runs 500 MMLU questions. No threshold is corrected for the nulls and scoring rules tried.

Before an ensemble's errors are read as missing knowledge, each sampled answer needs its own seed. The options should be rotated, a question counted correct only if it is right in every rotation, and any remaining structure tested against a fitted one-skill model and checked for near-duplicate models.

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A. Data and scoring

This appendix describes the questions, prompts and scoring, and checks five features of the data.

Questions. MMLU: ten random questions from each of the first 50 subjects in alphabetical order. HellaSwag: 200 validation items, sampled within each activity label. GSM8K: the first 100 test problems. In the shared-prompt runs, each pass draws five exemplars from the 285 MMLU development questions, so two passes can share an exemplar, and shows them, with their answer letters, before every question of the pass; both runs use the same exemplar sets.

Prompts and decoding. Each model receives one user message in its own chat template (raw text for Phi-2, which has none), checked against an independent reference tokenization. Generation uses vLLM 0.23.0 in bfloat16, with at most 8 new tokens for MMLU and HellaSwag and 256 or 1024 for GSM8K; the within-model runs sample at temperature T without top-k or top-p truncation. Every answer (pass k, question i) has its own seed, except in the shared-seed run, and vLLM gives each seeded request its own random-number generator, so an answer's noise does not depend on its position in a batch. The 24-model runs decode greedily with vLLM's deterministic engine settings; generation still differs slightly between engine starts.

Scoring. A reply counts as letter X if it begins with X (optionally after "Answer:", a parenthesis or bold markers) without negating or hedging, or if its first line says "the answer is X" or an equivalent. Any other reply, including one that names several letters, is unparseable and scored wrong. A GSM8K reply is correct when the number after its last "####", or else its last number, equals the gold answer.

Unparseable replies. Unparseable replies are rare within one model, at most 1.65% of a run's replies (Llama-3-8B on HellaSwag), and change no MP count: in the eight runs that have any, dropping every question with one, or imputing each such reply by the question's mean, leaves the MP count at zero in the seven independent-draw runs and at two in Llama-3-8B's shared-prompt run. Across models they are more frequent: in rotation 0, Mistral-7B v0.2 has 12.4% and every other model at most 9.6%, and 217 of the 500 questions have one in some rotation (Table A2).

Truncation on GSM8K. Truncation changes which answers are correct but does not couple two questions of a pass, so a truncated run still tests independence. At 256 new tokens, 55% of Qwen2.5-7B's GSM8K replies are cut off, and its accuracy is 51.0%, against 90.3% at 1024 tokens.

Replicates. GPU nondeterminism changes the ensemble little, but can move a result at a threshold. We collected the ensemble twice; the paper reports replicate A. Replicate B differs in 6 of the 48,000 chosen options and in eigenvalues by at most 0.081, and its all-4-correct λ_4, 20.82, lies just above the 2PL rank-4 99th percentile of 20.78 (1000 draws).

Near ties. A *cell* is one model's answer to one question in one rotation. It is a *near tie* when its two most probable first-token letters differ by less than 0.13 nats in log-probability in either replicate; 5.0% of cells are. Replacing each near tie by the majority correctness over the four rotations changes no number of ranks beyond skill (App. D), in rotation 0 or on all-4-correct. Dropping instead the 297 questions with a rotation-0 near tie leaves 203; λ_2 then exceeds the 3PL null but not the letter-aware null, both adequate.

Option references. Removing the 23 questions whose options refer to other options changes no number of ranks beyond skill in rotation 0 or on all-4-correct.

Table A1. The test on every within-model run ($\varepsilon = 0.05$). λ_+: MP edge. MC pct. z: Monte Carlo percentile of z (App. B) among 1000 simulated matrices of independent answers with the same pass rates. MP count: number of eigenvalues above λ_+. GSM8K runs use $T = 1$.

Run	N_b	λ_+	$\lambda_{\max}$	MC pct. z	**MP count**
Independent, MMLU ($K = 100, T = 1$)					
Qwen2.5-7B	43	2.74	2.56	60.1	0
Mistral-7B-v0.3	83	3.65	3.45	51.3	0
Llama-3-8B	173	5.36	4.97	10.2	0
SmolLM2-1.7B	406	9.09	8.74	28.6	0
Phi-3-mini	271	7.00	6.91	83.5	0
Independent, HellaSwag ($K = 50, T = 1$)					
Qwen2.5-7B	16	2.45	2.19	69.9	0
Mistral-7B-v0.3	65	4.58	4.05	19.7	0
Llama-3-8B	86	5.34	4.75	11.3	0
Independent, Qwen2.5-7B ($K = 30$)					
GSM8K, 256 tok.	62	5.94	5.32	37.9	0
GSM8K, 1024 tok.	19	3.22	2.50	12.0	0
MMLU, $T = 0.5$	25	3.66	3.39	78.5	0
MMLU, $T = 1.5$	85	7.20	6.53	32.7	0
Shared seed or prompt, MMLU ($K = 100, T = 1$)					
Qwen, seed	41	2.69	10.00	100	4
Qwen, prompt	93	3.86	4.77	100	2
Llama, prompt	264	6.89	11.31	100	2

B. The within-model test

This appendix explains the statistic of the within-model test, checks its calibration on every run, and measures its power.

The statistic. An eigenvalue test pools many weak correlations. If n questions all have pairwise correlation κ, the largest eigenvalue of their true correlation matrix is $1 + (n-1)\kappa$, which grows with n. A test of that eigenvalue can detect a correlation smaller than the noise of one sample correlation, about $1/\sqrt{K}$ for K passes. Each simulated matrix re-applies the borderline filter, so it has its own N_b and edge λ_+. To put all matrices on one scale we use $z = (\lambda_{\max} - \lambda_+)/\sigma_{\mathrm{TW}}$, with $\sigma_{\mathrm{TW}} = (1 + \sqrt{\gamma})(1 + 1/\sqrt{\gamma})^{1/3} K^{-2/3}$, the Tracy–Widom scale of $\lambda_{\max}$ for Gaussian data (Johnstone, 2001). For correlation matrices the limit is proved, as far as we know, only without centering or for Gaussian entries (Bao et al., 2012; Pillai & Yin, 2012). Our columns are centered Bernoulli variables with unequal pass rates, so we simulate the null distribution of z.

B.1. Calibration

The null. The null keeps each question's pass rate and draws answers independently: given the model and the prompts, each entry c_{ki} is a Bernoulli variable with $\Pr(c_{ki} = 1) = f_i$, the question's observed pass rate, independent across questions and passes. The borderline filter acts on each column alone, so the kept columns stay independent. A z above every draw of its null, as in the three shared-seed and shared-prompt runs, rejects independence but does not say whether the dependence lies between questions or between passes.

Number correct per pass. The number of correct borderline answers per pass varies as independent draws predict: its observed variance is 0.61 to 1.09 times the predicted one, and no run is overdispersed against a null that shuffles each column across passes (smallest $p = 0.27$). The two lowest ratios have underdispersion $p = 0.039$ and 0.045, not significant after a Holm correction over 48 tests (12 runs; over- and underdispersion, and trends over the pass index in the number correct and the top-eigenvector score). This check cannot detect a dependence that cancels in the number correct, such as the shared seed's (App. C).

Borderline threshold. The verdicts do not depend on the borderline threshold. At $\varepsilon = 0.01$, 0.02, 0.05, 0.10, 0.15 and 0.20, with the filter re-applied to the data and to each of the 1000 null matrices, no independent-draw run has an eigenvalue above its edge or a z above the 95th percentile of its null, and every shared-seed or shared-prompt run keeps at least two eigenvalues above its edge. The six question sets of a run are nested, so its six tests are not independent; at $\varepsilon = 0.20$ the three smallest sets have 6, 7 and 10 questions.

B.2. Power

Simulation. We plant a factor of known strength: one random number per pass, shared by a set of loaded questions. The matrices have the pass rates of two independent-draw MMLU runs, Qwen2.5-7B ($N_b = 43$) and Llama-3-8B ($N_b = 173$), $K = 100$ passes and all 500 questions. Entry c_{ki} is 1 when a hidden standard normal score falls below $\Phi^{-1}(f_i)$, so each question keeps its pass rate. On loaded questions the score is $\sqrt{\rho}\,u_k + \sqrt{1-\rho}\,e_{ki}$, with u_k shared by the loaded questions of pass k and e_{ki} independent noise; elsewhere it is e_{ki}. Two loaded questions then correlate at ρ, the latent correlation, in their scores and, at these pass rates and averaged over pairs, at about $\rho/2$ in their 0/1 entries. The factor loads on all borderline questions (diffuse), on 30 at random, or on 9 chosen at random or from one subject. It is detected when the MP count, with the filter re-applied, is positive. Each scenario has 500 matrices (2000 at $\rho = 0$).

Results. The test detects diffuse structure and misses small clusters. A diffuse factor with $\rho = 0.05$ is detected in every simulation at Llama-3-8B's 173 borderline questions and in 47% at Qwen2.5-7B's 43, because at 173 it loads four times as many questions (Fig. A1). A cluster of fixed size is harder to detect at 173, where it is a smaller share of the borderline questions: a 9-question cluster at $\rho = 0.2$, the largest strength simulated, is detected in 23% of Qwen2.5-

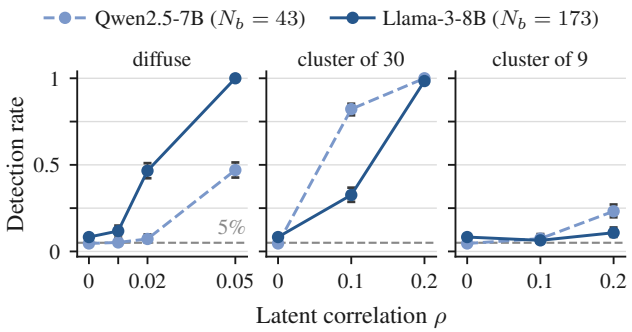


Figure A1. Detection rate of the MP count against the latent correlation ρ of a factor planted on all borderline questions or on 30 or 9 of them, at the pass rates of two MMLU runs. Whiskers: Wilson 95% intervals.

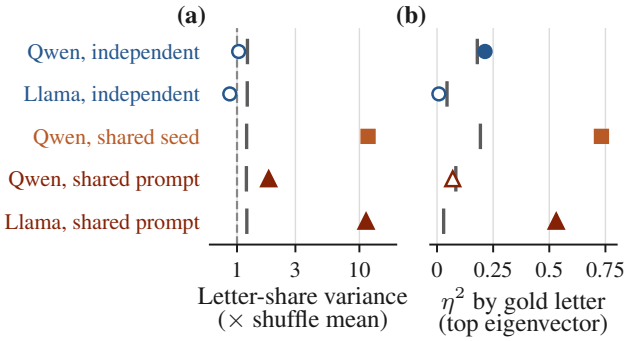


Figure A2. A shared seed or prompt makes a pass's letter shares vary beyond chance. Of the shared runs, only the seed run and Llama-3-8B's prompt run also group the top eigenvector's loadings by gold letter. **(a)** Across-pass variance of a pass's letter shares, summed over letters, divided by the mean of that sum after each question's answers are shuffled across passes. **(b)** η^2, the share of the variance of the top eigenvector's loadings that the gold letter explains. Grey tick: null 99th (a) or 95th (b) percentile; filled: beyond it.

7B and 11% of Llama-3-8B simulations when chosen at random, and in 24% and 12% when drawn from one subject. Without a factor, the MP count has false-positive rates of 4.6% and 8.3%. We did not simulate other values of N_b.

Subject structure in the data. Borderline questions of one subject correlate no more than other pairs: mean within-subject correlations are -0.018 ($p = 0.80$) for Qwen2.5-7B and 0.002 ($p = 0.37$) for Llama-3-8B (MMLU, $T = 1$).

C. What a shared seed or prompt couples

This appendix shows how a shared seed and a shared prompt couple the questions of a pass.

Shared noise. A shared seed gives every question of a pass the same random numbers at every step, whatever its prompt. At each step, a seeded request draws one exponential random number per token and emits the token whose probability at temperature T, divided by that number, is largest. This is equivalent to Gumbel-max sampling, so each answer alone follows the model's distribution. Coupled generation and counterfactual sampling use it on purpose (Corvelo Benz et al., 2026; Ravfogel et al., 2025).

The shared-seed run. The shared seed moves all answers of a pass toward the same letters: in Qwen2.5-7B's shared-seed run, the across-pass variance of a pass's letter shares (its fractions of A, B, C and D answers), summed over letters, is 11.8 times its mean after each question's answers are shuffled across passes, against 1.03 in the independent-draw run (Fig. A2a). A pass whose noise favors one letter gains on questions with that gold letter and loses on the others, so the top eigenvector's loadings group by gold letter (Sec. 3.1), while the gains and losses cancel in the number correct per pass (observed over predicted variance 0.83).

The shared-prompt runs. Llama-3-8B copies the exemplars' letters, as few-shot prompts are known to induce (Zhao et al., 2021): the number of a pass's exemplars whose answer is letter L correlates with the pass's share of answers L at $r = 0.35$ to 0.47. Its letter shares vary 11.3 times their shuffled mean, and its number correct per pass is overdispersed (ratio 1.51, $p = 0.001$). Qwen2.5-7B does not copy the letters: its letter shares vary 1.8 times their shuffled mean and do not follow the exemplars' letters ($|r| \leq 0.09$; loadings by gold letter $\eta^2 = 0.07$). Instead, its passes that share an exemplar agree more on borderline questions than passes that share none, by 10.7 standard deviations of a null that permutes the exemplar sets among passes (5.9 for Llama-3-8B; $p = 0.0005$ for both).

Gold letter below the edge. In Qwen2.5-7B's independent-draw run, whose $\lambda_{\max}$ lies below λ_+, the top eigenvector's loadings group by gold letter with $\eta^2 = 0.21$ ($p = 0.023$; Fig. A2b). This is one uncorrected test on an eigenvector inside the noise range, not interpreted.

D. The 24-model ensemble

This appendix defines the models, nulls and tests, and gives the evidence behind Secs. 3.2 and 3.3.

D.1. Models, nulls and tests

Roster. Five Qwen (Qwen2.5 at 0.5B, 1.5B, 3B and 7B; Qwen2-7B), five Mistral (Mistral-7B v0.2 and v0.3, Zephyr-7B-beta, Ministral-8B, Mistral-Small-22B), four Llama (Llama-3.2 at 1B and 3B, Llama-3-8B, Llama-3.1-8B), three Phi (Phi-2, Phi-3-mini, Phi-3.5-mini), two Gemma-

2 (2B, 9B), and SmolLM2-1.7B, OLMo-2-7B, Yi-1.5-6B, DeepSeek-LLM-7B and Granite-3.1-8B. Phi-2 is a base model; the other 23 are instruction-tuned. Rotation-0 accuracy runs from 35.0% (Llama-3.2-1B) to 72.8% (Gemma-2-9B).

One-skill nulls. In every null, skill is the only link between questions. Every simulated draw is conditioned on each kept column staying split. The 2PL (Sec. 2) treats the skills as fixed and integrates the question parameters over a normal population, by marginal maximum likelihood; each simulated column draws its (a_i, b_i) from its posterior in that fit, made on all 500 questions. Rasch is fitted by joint maximum likelihood. In KG-u, a model either knows a question, with the 2PL probability, in all four rotations at once, or guesses a letter uniformly in each rotation; KG-u is fitted to all four rotations and scored like the data. The letter know-or-guess null (KG-l) instead draws guesses from the model's letter preference π_m, its probabilities of guessing A, B, C and D. We estimate π_m from each model's letter shares and accuracy over all four rotations, assuming that a known answer falls on each letter equally often across rotations and a guess falls on letter L with probability $\pi_m(L)$; for Llama-3.2-1B, $\pi_m(\mathrm{A}) = 0.90$. The letter-aware null uses the same π_m but is fitted to rotation 0 only: model m is correct with probability $\pi_m(g_i) + (1 - \pi_m(g_i))\,\sigma(a_i(\theta_m - b_i))$, where g_i is the gold letter. The soft-preference (SP) null is defined in App. D.2. We also tried a Bock–Aitkin 2PL, with skills integrated and slopes at most 4, and a fixed-margin null, uniform over 0/1 matrices with the observed margins.

Adequacy. The adequate nulls differ between the two scores. In rotation 0 (1000 draws per null) the observed λ_1 lies at the 76.4th percentile of the 3PL, 86.7th of the letter-aware, 98.7th of the KG-u and 85.7th of the SP null, which are adequate, and at the 100th of the 2PL and Rasch. On all-4-correct (2000 draws) it lies at the 31.5th and 54.2nd percentiles of the 2PL and Rasch, which are adequate, at the 100th of the 3PL and at the 0th of the KG-u and SP. KG-l, Bock–Aitkin and fixed-margin are inadequate for both.

Max-statistic test. The test uses 2000 new null draws. Its statistic M is the largest of the percentiles of λ_2, λ_3 and λ_4 among these draws, ties counted as half. Each draw's M_d is computed likewise against the other draws, and $p_M = (1+\#\{d : M_d \geq M\})/2001$; the test rejects at $p_M < 0.05$. Three null draws reach $M = 100$, so p_M cannot fall below 0.002, the value in rotation 0. The *rank-2 rule* rejects when λ_2 exceeds the null's 99th percentile. For an adequate null, the *number of ranks beyond skill* counts ranks $2, 3, \ldots$ above their 99th percentiles, stopping at the first that is not.

Order of analyses. The max-statistic test was written down before its first run, together with the list of sibling groups, the adequacy rule, a calibration under the SP null and a power study; the adequacy rule itself had been set after the first analyses. These choices still depend on earlier analyses: an earlier post hoc joint statistic had already rejected, and the separate rank tests and the correlations between model pairs had already been computed. Removing single models or pairs, and the reported analyses of test size, power and the difficulty factor, were decided after seeing the results.

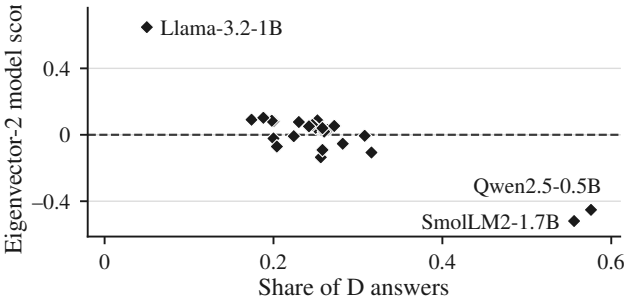


Figure A3. Each model's rotation-0 score on eigenvector 2 of the 24-model matrix against its share of D answers.

Size. Under a true one-skill model the test rejects slightly too often, and the adequacy rule can pick the wrong null. The size of a test is its false-positive rate when the null model is true. We simulate 300 all-4-correct datasets from each fitted one-skill model, 2PL and Rasch, refit both nulls to each (500 draws), and apply the adequacy rule and the max-statistic test (Table A2). The 2PL test, the max-statistic test against the 2PL null, rejects in 6.7% of datasets under 2PL truth and 8.7% under Rasch truth (Clopper–Pearson 95% intervals 4.1–10.1% and 5.7–12.4%), so under Rasch truth it exceeds the bound of 7.5% we set before the run. The Rasch test rarely rejects under Rasch truth (0.7%), mainly because the refitted Rasch null is adequate in only 23% of those datasets. The adequacy rule accepts the Rasch null on every 2PL dataset, although its slopes are wrong there, and rejects the correct Rasch null in most datasets drawn from it. The calibrated p_M is the fraction of the 300 datasets from one truth whose p_M is at most the observed one.

D.2. Rotation 0: the letter eigenvector

Letter evidence. Eigenvector 2 follows the gold letter: its loadings group by gold letter ($\eta^2 = 0.48$, $p = 0.0002$), and the models' scores on it track their share of D answers (permutation $p = 10^{-4}$; Fig. A3). The loadings on eigenvector 1 also group by gold letter ($\eta^2 = 0.22$), and those on eigenvectors 3 and 4 more weakly, though beyond chance ($\eta^2 = 0.087$ and 0.061, $p = 0.0002$ each). Because gold letters are nearly balanced (118 to 129 per letter), a letter preference barely changes expected benchmark scores.

Where the letter preference acts. The letter-aware null lets the preference act only on guesses; the 3PL and KG-u nulls have none. A preference that also tilts answers a model partly knows can, with one skill, put λ_2 above the letter-aware and 3PL nulls: in 200 datasets from the SP fit, each null refitted (200 draws), λ_2 exceeds the letter-aware null's 99th percentile in 60% and the 3PL null's in 99.5%.

The soft-preference null. Model m chooses option j of question i in a rotation with probability proportional to $\exp\big(b_{ij} + \mathbb{1}[j = g_i](a_i\theta_m + e_{mi})\big)\,\pi_m(L_j)$. Here g_i is the correct option, b_{ij} is a fitted intercept for option j with $b_{ig_i} = 0$, L_j is the letter at which the rotation places option j, and $e_{mi} \sim N(0, \tau^2)$ is shared by the four rotations of one model on one question and independent across models. Skill raises only the correct option, while π_m multiplies every option. The fit is penalized maximum likelihood on the parsed choices of all four rotations ($\tau = 1.17$).

Calibration of the SP null. Against the calibrated SP null, λ_2 alone is unlikely ($p = 0.010$), but the max-statistic test does not reject ($p_M = 0.079$); neither verdict is fully validated. Simulations drop the cells unparseable in the data: without that mask, the observed λ_2 lies at the 99.8th percentile of the null; with it, at the 95.1st. Refitting inflates the slopes, so tail probabilities are recalibrated on 200 datasets from the fit, each with its own null of 1000 draws. The calibrated p of rank j is the fraction of those datasets whose percentile for λ_j is at least the observed one. For λ_2 in rotation 0 it has Monte Carlo standard error 0.007, and is 0.015 under the convention $(1+k)/(1+n)$. The calibration is exact only if each refit inflates the slopes as much as the first, and does not correct the misfit described next. The null is inadequate without Llama-3.2-1B and on all-4-correct.

Predictive check. The SP null fails the predictive check we set before the run, against 500 datasets simulated from the fit: it reproduces per-model accuracy and letter shares but not the number of rotations in which a model answers a question correctly. Of the shares of questions with each count from 0 to 4, per model, only 82.5% lie inside the central 98% of their simulated values (90% required), and the observed number of split questions under "correct in at least two rotations" lies above all simulations. The weakest, letter-biased models have too many questions correct in exactly one rotation, and 21 of 24 models too many in all four (largest excess 0.06 of the questions).

Without Llama-3.2-1B (post hoc). Without Llama-3.2-1B, the rotation-0 excess moves from λ_2 to λ_4, which exceeds the 99th percentiles by small margins. The λ_2 of the other 23 models (Sec. 3.2) lies below the 3PL null's 99th percentile, 40.1. The max-statistic test rejects the 3PL null ($p_M = 0.003$; Sec. 3.2) and the letter-aware null ($p_M = 0.015$), both through λ_4. Ranks 3 and 4 lie at the 98.5th to 99.9th percentiles of the 3PL and letter-aware nulls, and in their 1000 draws λ_4 exceeds the 99th percentile by only 0.35 (3PL) and 0.57 (letter-aware).

Unparseable replies and sibling models. Without the 112 questions with an unparseable rotation-0 reply, the max-statistic test still rejects the letter-aware null ($p_M = 0.013$) and the 3PL ($p_M = 0.004$), although λ_2 falls to the 98.3rd percentile of the letter-aware null, below its 99th, while it stays above the 3PL null's 99th. Keeping one model from each sibling group (Sec. 3.3), in any of the 24 ways, leaves both nulls adequate, and the test rejects both in every subset.

Option position and letter preference. Part of a model's self-agreement depends on option position (Ding, 2026). In the option-level response model of Fei et al. (2026), a per-model positional preference adds little to the prediction of held-out correctness (accuracy 0.840 with it, 0.838 without). We did not test whether letter preference affects item-level factor and item response analyses (Kipnis et al., 2025; Sha & Zhao, 2026; Yao et al., 2026), or analyses of shared wrong answers (Kim et al., 2025), as it affects these eigenvectors.

D.3. All-4-correct: the marginal excess

Power. The test detects a second skill as strong as the first, but usually not one half as strong. We simulate from the adequate all-4-correct 2PL fit plus a second skill. As in the 2PL null, each of the 391 kept columns draws (a_i, b_i) from its posterior. On the 125 of them from the 18 STEM subjects, the logit $a_i(\theta_m - b_i)$ gains $s\,\bar{a}\,\mathrm{sd}(\theta)\,z_m$. Here $z_m \sim N(0,1)$ is model m's second skill, independent of θ_m. The factor $\bar{a} = 1.71$ is the median slope, and $\mathrm{sd}(\theta) = 1$, so at $s = 1$ the second skill matches the first in spread and median slope. We refit both nulls to each of 200 datasets per s (500 draws; rates in Table A2). At $s = 1$ the rank-2 rule detects it in 69–78% of datasets; at $s = 2$ the max-statistic test and the rank-2 rule detect it in at least 97%. The median λ_2 is 28.2 at $s = 0$, 28.3 at $s = 0.5$ and 33.8 at $s = 1$, against the observed 29.0.

The difficulty factor. On all-4-correct, eigenvector 2 behaves like the difficulty factor that the one-skill nulls produce: its loadings follow each question's accuracy across models, the share of the 24 models answering it correctly, which runs opposite to difficulty. The observed $|r|$ between loading and accuracy, 0.57, lies just below the 2PL null's 5th percentile and inside the Rasch null's 5–95% range. In the rotation-0 3PL null the loadings barely follow accuracy (median $|r|$ 0.09; observed 0.04). The observed all-4-correct loadings group by subject ($\eta^2 = 0.21$, $p = 0.001$), but a cubic in accuracy explains 39% of their variance, and after its removal the subject grouping is at chance ($p = 0.24$).

Table A2. Max-statistic test on all-4-correct (reject at $p_M < 0.05$), against the 2PL and Rasch nulls. Pct.: percentile of each eigenvalue in the null (2000 draws); under the 2PL, λ_4 lies at the 99.1st, just below the 99th percentile in the 1000 draws of Fig. 2c. Subsets: number that reject, and the range or maximum of p_M; sibling: one model kept from each sibling group; random: five of the other 15 models removed. Pair removals: one Phi and one Llama model removed. Two other models removed: the 190 removals of two models outside the pairs (fraction that reject, maximum p_M). Unparseable: replies imputed, or questions dropped (1000 draws), either the 95 on which some model has no wrong parsed rotation and at least one unparseable one, or all 217 that contain one. Size: rejection rate in 300 datasets simulated from each fitted one-skill model. Calibrated p_M: under 2PL and Rasch truth. Power: max-statistic test / rank-2 rule, 200 datasets per strength s (2PL fit).

	2PL	**Rasch**
Pct. of $\lambda_2, \lambda_3, \lambda_4$	71.8, 98.5, 99.1	94.6, 97.0, 98.6
p_M, 24 models	0.026	0.041
Sibling subsets	0/24 (0.09–0.65)	0/24 (0.12–0.69)
Random subsets	24/24 ($\leq$0.047)	24/24 ($\leq$0.046)
Pair removals	0/4 (0.22–0.69)	0/4 (0.32–0.38)
Two other models removed	0.90 ($\leq$0.099)	0.74 ($\leq$0.133)
Unparseable: impute	0.019	0.042
drop 95	0.036	0.043
drop 217	0.23	0.12
Size, 2PL truth	0.067	0.067
Size, Rasch truth	0.087	0.007
Calibrated p_M	0.033, 0.047	0.057, 0.017
Power, $s = 0.5$	0.20 / 0.01	0.21 / 0.07
Power, $s = 1$	0.91 / 0.69	0.93 / 0.78

Sibling models. Keeping one model per sibling group (Sec. 3.3), in any of the $2 \cdot 2 \cdot 3 \cdot 2 = 24$ ways, ends the rejection: in each subset of 19 models both refitted nulls (1000 draws) are adequate, and the test rejects in none. In 24 control subsets that remove five of the other 15 models at random, it rejects in all (Table A2). The subsets overlap, so this is one contrast, not 24 replicates. The removed models are not size-matched to the sibling models, and the 15 include models from the same developers (Phi-2, Qwen2.5 0.5B–3B, Llama-3.2, Ministral-8B, Mistral-Small).

The two pairs (post hoc). The excess rests on the Phi and Llama pairs (Sec. 3.3): removing one model from each, in all four ways, gives a larger p_M under both nulls than any of the 190 removals of two other models, of which 90% reject under the 2PL and 74% under Rasch (1000 draws; every null adequate; Table A2). Three other candidates do not carry the excess. The Qwen pair: removals that include Qwen2-7B or Qwen2.5-7B reject more strongly, with mean p_M 0.010 and 0.006 under the 2PL. The Mistral group: removing Mistral-7B v0.2 and v0.3 (third-largest residual correlation) leaves $p_M = 0.019$, and its other two removals give p_M of 0.004–0.023. Accuracy: the 15 removals of two models with rotation-0 accuracy 0.62–0.71, around the pair models' 0.644–0.684, give p_M at most 0.052 (2PL) and 0.113 (Rasch), below every pair removal.

Table A3. Simple baselines. Top: rotation-0 letter shares of the three models whose largest share exceeds 0.5; no other model's largest share exceeds 0.35. Bottom: correlation of model pairs' residuals after the all-4-correct 2PL fit, with rank among 276 pairs (mean -0.04, standard deviation 0.08).

Model	A	B	C	D
Llama-3.2-1B	0.81	0.04	0.09	0.05
Qwen2.5-0.5B	0.23	0.06	0.12	0.58
SmolLM2-1.7B	0.07	0.20	0.17	0.56

Model pair	r	**Rank**
Phi-3-mini, Phi-3.5-mini	0.40	1
Llama-3-8B, Llama-3.1-8B	0.27	2
Mistral-7B v0.2, v0.3	0.20	3
Mistral-7B v0.3, Llama-3-8B	0.19	4
Mistral-7B v0.2, Zephyr-7B	0.17	5
Qwen2-7B, Qwen2.5-7B	0.03	37

Model families (post hoc). The models' scores on all-4-correct eigenvector 4, whose eigenvalue lies below λ_+, group by family ($\eta^2 = 0.74$, $p = 0.006$; Holm-corrected over the four eigenvectors tested, $p = 0.024$).

Llama-3.2-1B (post hoc). Llama-3.2-1B carries part, not all, of the excess. Removing it alone makes the KG-u null adequate ($p_M = 0.15$). The sum $\lambda_2 + \lambda_3 + \lambda_4$, specified with the max-statistic test and not used for decisions, gives $p = 0.023$ (2PL) and 0.016 (Rasch). Of the 190 removals, 12 of the 19 that include it still reject under the 2PL and 6 under Rasch. Removing it and one model per pair gives p_M of 0.30–0.59 (2PL) and 0.38–0.60 (Rasch).

Simple baselines. Rotation-0 letter shares, which need no model fit, flag three models whose most frequent letter exceeds half of their replies (Table A3). These shares count unparseable replies; $\pi_m(\mathrm{A}) = 0.90$ of Llama-3.2-1B counts only guesses. A residual is a model's observed correctness minus its 2PL fitted probability, over the 391 split all-4-correct questions. Residual correlations between models rank the Phi and Llama pairs first and second of 276.

Other scores. Counting a question correct in at least one, two or three rotations, or by the mean over rotations, supports no conclusion. At "at least two" the 2PL null is rejected ($p_M = 0.002$) but the 3PL and SP nulls are not ($p_M = 0.86$ and 0.85); at "at least three" Rasch, KG-l and the 2PL are rejected (p_M 0.008–0.010) but KG-u is not ($p_M = 0.28$). These rejections mean little: in datasets from the one-skill SP fit, a refitted 2PL null is rejected in 98.5% at "at least two", 58% at "at least three". "At least one" gives $p_M = 0.996$ (SP null); the mean score has no adequate null.